

Mohak Kapoor

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SUMMARY

AI Engineer (B.Tech, 2026) building generative AI and applied ML systems end-to-end: LLM integration, retrieval pipelines, agentic architectures, and deployment. Designed and shipped a multi-repo RAG assistant with a custom hybrid retrieval architecture and an LLM-as-judge evaluation framework built from scratch, and engineered a two-stage hybrid ML detection system deployed to edge hardware via a Dockerized FastAPI inference service. Comfortable across the full stack: data pipelines, model/system development, quantitative evaluation, and production deployment.

PROJECTS

CodeAtlas : Multi-Repo Agentic RAG Assistant | LangChain, Qdrant, Gemini [\(Live Link, GitHub Link\)](#) *Present*

- Increased context recall from 0.48 to 0.83 on a 52-case benchmark by building a hybrid retrieval pipeline that combines dense search with BM25 and cross-encoder reranking, replacing a naive single-repo RAG baseline with a system across an entire GitHub portfolio.
- Indexed code and docs into a Qdrant vector database using local Hugging Face embeddings, then wired up a LangChain-orchestrated chatbot over Gemini so answers came back grounded and citation-backed instead of hallucinated.
- Designed JudgeKit, a dual-agent LLM-as-judge evaluation framework, to score retrieval and generation quality; those scores surfaced weak spots in prompt design that I iterated on through three versions of the retrieval architecture.

Network IDS Detection Engine | Autoencoder, PyTorch, FastAPI, Docker [\(Live Link, GitHub Link\)](#) *Jan–May 2026*

- Reached 92.4% detection accuracy across four benchmarked architectures (XGBoost, PyTorch feed-forward network, SVM) by engineering a two-stage hybrid anomaly-detection pipeline on 2.8M rows of network traffic data, using unsupervised models to flag candidates before supervised classification.
- Cut the feature space 51% while retaining 98.97% of variance by applying incremental PCA, after building an end-to-end data pipeline for ingestion, EDA, preprocessing, and temporal train/test splitting to prevent leakage across unseen attack types.
- Shipped the model as a Dockerized FastAPI inference service to Raspberry Pi, exposing predictions through a REST API secured with bearer-token auth over a Cloudflare Tunnel, with a live WebSocket dashboard streaming real-time latency and metrics.

EXPERIENCE

HumanizeIQ

AI Intern

Jun 2025 – Sep 2025

Hazlet, New Jersey (Remote)

- Designed an n8n-orchestrated LLM evaluation pipeline that scored recruiter call performance, writing the evaluation logic itself to check script adherence against the transcript, recruiter identity, and job role via prompt-based scoring.
- Restructured an existing Postgres call-transcript database into normalized calls and follow-back tables, then built a FastAPI backend with REST endpoints for transcript, identity, and script retrieval plus endpoints to persist scores, cutting analyst turnaround from two hours to under five minutes.
- Built two production MCP servers in TypeScript on Cloudflare Workers to give an LLM agent tool-calling access to internal services: an image-generation MCP calling an internal LLM gateway with outputs persisted to Cloudflare R2, and a document-orchestration MCP composing multi-modal HTML reports by invoking the first MCP as a tool.
- Led daily Scrum stand-ups in an agile environment for the full internship, and independently built the AnythingLLM orchestrator integration for both MCP servers, enabling agent-driven document and image generation workflows.

CERTIFICATIONS & ACHIEVEMENTS

- JPMorgan Chase & Co. - Quantitative Research Job Simulation, Forage [\(Certificate\)](#)
- Machine Learning Specialization, DeepLearning.AI, Stanford University [\(Certificate\)](#)
- LeetCode Knight (Contest Rating 2113), including a rank of 73/38291 at Biweekly Contest 190 [\(Leetcode Profile\)](#)
- Winner, OSDHack'23 (JIIT)

TECHNICAL SKILLS

LLMs, GenAI & Retrieval	LLM Integration (Gemini, OpenAI GPT-4), Prompt Engineering, RAG, LangChain, Agentic Systems, Agentic Tool Calling, MCP, Transformers, ChromaDB, Qdrant, Vector Databases, Hugging Face Embeddings, Hybrid Retrieval, Retrieval Eval.
ML & Modeling	Supervised Learning, XGBoost, CNN, Anomaly Detection, Time-Series Forecasting, Statistical Modeling, Feature Engineering, Cross-Validation, Hyperparameter Tuning
MLOps & Deployment	FastAPI, REST APIs, Microservices, Docker, Edge Deployment (Raspberry Pi 5), CI/CD, AWS
Languages	Python, JavaScript, TypeScript, C++, SQL
Frontend & Full-Stack	React, Next.js, Node.js, JSX, Vue
Systems & Platform	OOP, Linux, Data Structures & Algorithms, Agile/Scrum, SDLC
Frameworks & Data	TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, PostgreSQL, SQL, NoSQL, Git

EDUCATION

Jaypee Institute of Information Technology

B.Tech in Electronics and Communication Engineering

Noida, India

2022-2026

- Relevant Coursework:** Machine Learning (A+), Deep Learning (A+), Advanced Statistics (A), Queueing Theory, Biomedical Signal Processing, Affective Computing, Computer Systems, Multivariate & Differential Calculus, Control Systems